ANTIPSYCHOTICS

Dr. Shanti Gurung

Learning Objectives

Understand: Describe schizophrenia and explain its neurochemical basis and theories.

Understand: Explain the dopaminergic and serotonergic mechanisms of antipsychotic drugs.

Analyze: Compare classical and atypical antipsychotics in treating positive and negative symptoms.

Remember: List clinical uses of antipsychotics other than schizophrenia.

Analyze: Differentiate the adverse-effect profiles of classical and atypical antipsychotics.

Understand: Describe the time course, signs, and symptoms of antipsychotic-induced extrapyramidal symptoms (EPS).

Psychotic illnesses (Psychoses)

Psychosis is a symptom of mental illnesses characterized by a distorted or non-existent sense of reality

Common Examples:

- Schizophrenia
- Manic phase of bipolar disorder
- Schizoaffective (both schizophrenic and mood disorders)
- OCD (Obsessive compulsive disorder)
- Psychosis induced by Alzheimer's or Parkinson's disease
- Substance-induced psychosis (CNS stimulants)

Schizophrenia


catatonic behavior

Schizophrenia is defined by the DSM-5:

As the presence of psychotic symptoms like

- Delusions
- Hallucinations
- Disorganized speech
- Grossly disorganized or catatonic behavior
- Diminished emotional expression or volition

For a significant portion of time during a one-month period (or less if successfully treated)

Diagnostic and Statistical Manual (DSM)

Clinical Presentation

Positive symptoms

- Delusions (are often paranoid: perceived threat from others)
- Hallucinations (auditory)
- Disordered thoughts (illogical & irrational)
- Hypersensitivity to environmental stimuli, with feelings of enhanced sensory awareness

Negative symptoms (apathy, avolition, alogia)

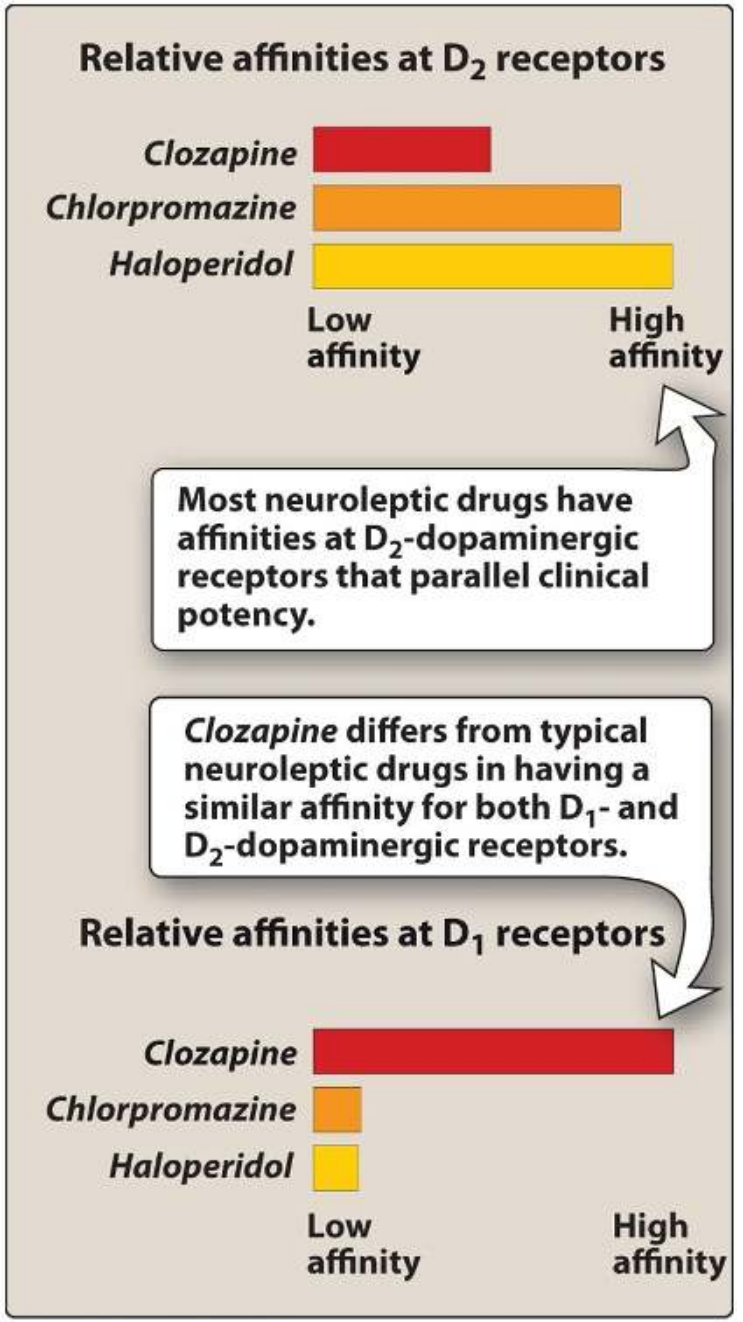
- Social withdrawal (often with deterioration in personal care, disturbed interpersonal relationships)
- Emotional withdrawal (depression in most patients)
- Impoverished speech (Neologisms -made-up words or phrases; Echolalia -repetition of others' words; Verbigeration - repetition of senseless words or phrases)

Cognitive symptoms

- Decreased working memory
- The ability to use information immediately after learning it
- Trouble focusing or paying attention

The Dopamine Hypothesis: The Evidence

- The observation that Dopamine **agonists** can induce psychosis (Amphetamine or Levodopa)
- The efficacy of the typical **Antipsychotics,** which strongly block **D_2 receptors**
- Increase in Dopamine **receptor density** in the brains of untreated patients with schizophrenia



Biological Basis of Schizophrenia - Dopamine Receptors

1. Mesolimbic System

   From the ventral tegmental area (VTA) to amygdala, nucleus accumbe involved in emotions, memory, motivational behavior

   Increase in DA causes positive symptoms of schizophrenia

2. Mesocortical system

   From VTA to prefrontal cortex, frontal cortex… involved in cognition, communication, social function

   DA Hypo activity contribute to negative and cognitive symptoms of schizophrenia

3. Nigrostriatal Pathway

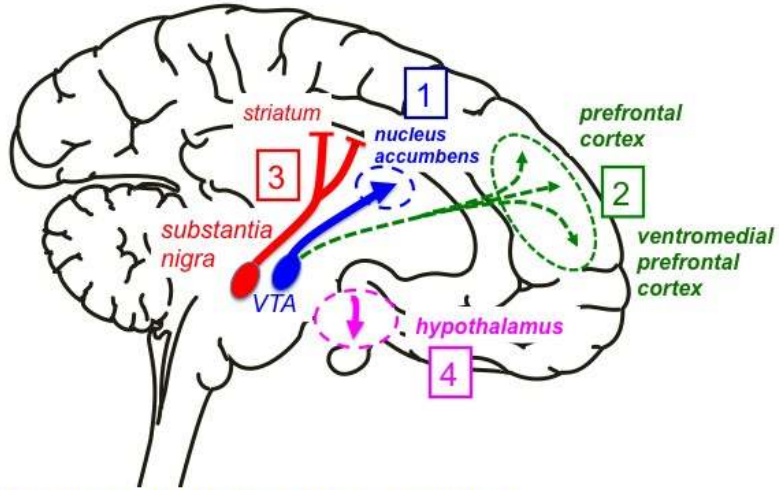
   From the substantia nigra to the striatum, which modulates the extrapyramidal motor system (posture, movement)

   Antipsychotic drugs cause EPS and TD side effects

4. Tuberoinfundibular Pathway

   From the hypothalamus to the pituitary gland, which modulates prolactin release

   Antipsychotic drugs cause hyperprolactinemia as side effect



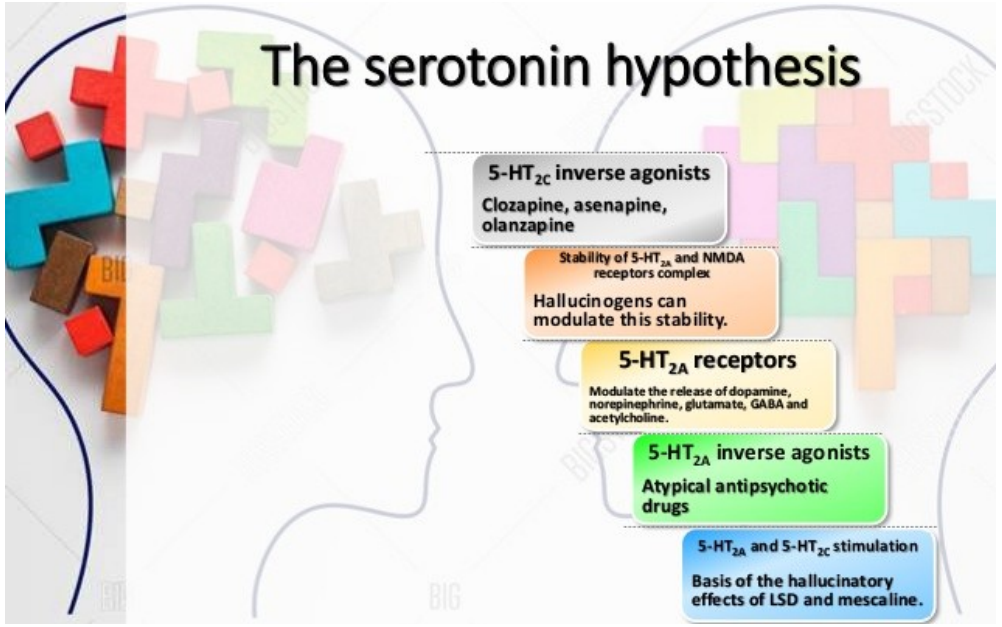
Problems with the Dopamine Hypothesis

- Antipsychotic drugs are only partially effective
- NMDA antagonists like **phencyclidine (PCP)** can produce more "schizophrenia-like" symptoms than dopamine agonists
- Newer (atypical) antipsychotics are less selective of the D_2 receptor

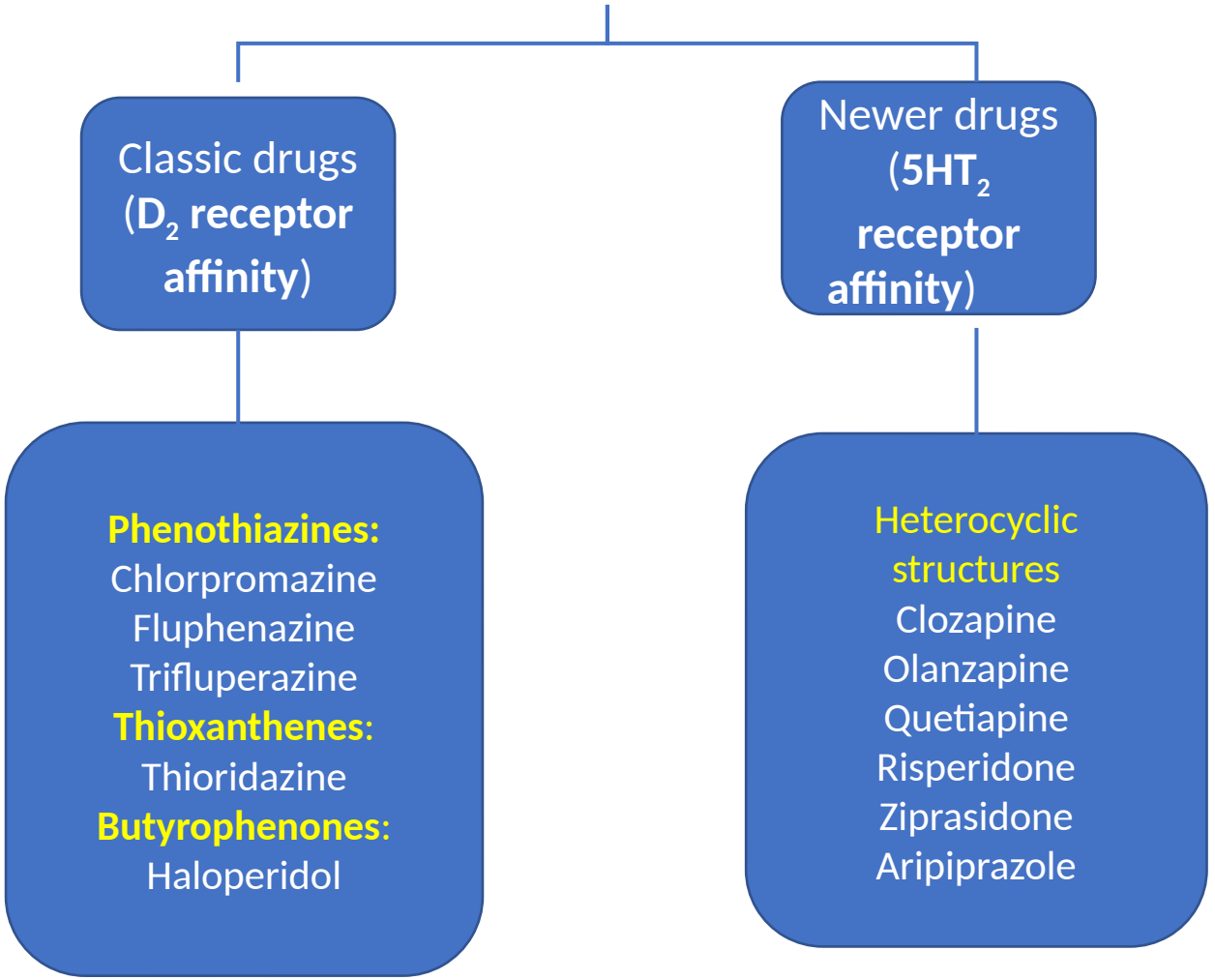
Newer (atypical) antipsychotics	Receptor affinity
Clozapine	D_4 << 5HT
Olanzapine	D_2 << $5HT_{2A}$
Ziprasidone	D_2<< $5HT_{2A}$ and $5HT_{1D}$
Aripiprazole (Abilify)	Partial dopamine (D2) agonist

The Serotonin Hypothesis: The Evidence

- New hypothesis suggests that serotonin (5-HT2) is also involved in the pathophysiology
- Drugs like LSD and Mescaline (serotonin agonists) cause hallucinations
- 5-HT2 blockade is the principal mechanism of atypical antipsychotic drugs like clozapine



ANTIPSYCHOTICS



Classical(Typical) antipsychotics

Mechanism: D_2 receptor blockade

Effects:

- Decrease positive symptoms (by blocking mesolimbic D_2)
- Cause **EPS (extrapyramidal symptoms)** (by blocking nigrostriatal D_2)
- Cause **hyperprolactinemia** (by blocking tuberoinfundibular D_2)

Classification:

High-potency (stronger D_2 blockade → more EPS, less sedation & anticholinergic effects)

- **Haloperidol**
- **Fluphenazine**
- **Trifluoperazine**

Low-potency (weaker D_2 blockade → more sedation & anticholinergic effects, less EPS)

- **Chlorpromazine**
- **Thioridazine**

New (Atypical) antipsychotics

Mechanism: Block **5-HT_2A , D_2** (some also block D4, α1, H1)

Effects:

- **Reduce positive symptoms** (D_2 blockade in mesolimbic)
- **Improve negative symptoms** (5-HT_2A blockade → ↑ dopamine in mesocortical)
- **Lower risk of EPS & hyperprolactinemia** (weaker nigrostriatal & tuberoinfundibular D_2 blockade)

COMPARISON TABLE OF ATYPICAL ANTIPSYCHOTICS BASED ON THEIR RECEPTOR PROFILES:

Drug	D2 Receptor	5-HT2A Receptor	5-HT1A Receptor	α1 Adrenergic	H1 Histamine	M1 Muscarinic	Notable Features
Clozapine	Antagonist	Strong Antagonist	Partial Agonist	Strong Blockade	Strong Blockade	Strong Blockade	**Agranulocytosis risk**, most effective for treatment-resistant schizophrenia
Olanzapine	Antagonist	Strong Antagonist	Weak Agonist	Moderate	Strong Blockade	Strong Blockade	High **metabolic syndrome risk**(weight gain, diabetes)
Quetiapine	Antagonist	Antagonist	Partial Agonist	Moderate	Strong Blockade	Weak Blockade	**Sedation**, used in bipolar depression
Risperidone	Antagonist	Strong Antagonist	Weak Partial Agonist	Strong Blockade	Moderate	Weak Blockade	High **prolactin elevation, dose-dependent EPS**
Paliperidone	Antagonist	Strong Antagonist	Weak Partial Agonist	Strong Blockade	Moderate	Weak Blockade	Active metabolite of **risperidone**, similar profile
Ziprasidone	Antagonist	Strong Antagonist	Strong Agonist	Moderate	Weak Blockade	No Effect	**QT prolongation risk**, least weight gain
Aripiprazole	**Partial Agonist**	Antagonist	**Strong Agonist**	Moderate	Weak Blockade	No Effect	**Unique MOA (D2 partial agonist), minimal metabolic side effects**

Side effects and clinical considerations

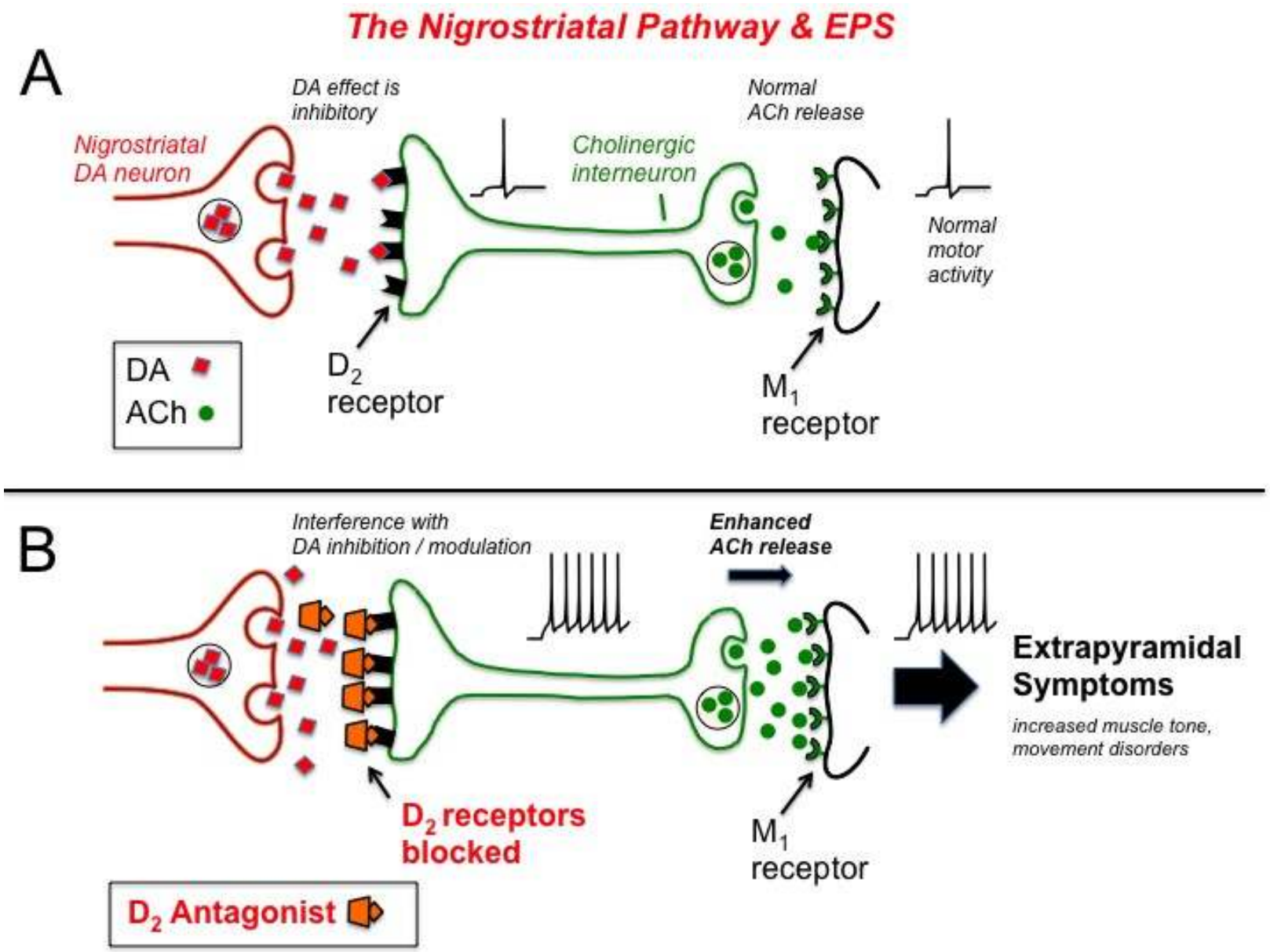
❖**Extrapyramidal Symptoms (EPS) – More with Typical Antipsychotics**

- **Acute dystonia** (hours-days) → Treat with **Benztropine**
- **Akathisia** (restlessness) → Treat with **Propranolol**
- **Parkinsonism** (weeks) → Treat with **Benztropine or Amantadine**
- **Tardive Dyskinesia** (months-years) → Treat with **Valbenazine**

❖**Other Adverse Effects**

- **Hyperprolactinemia** (Galactorrhea, Amenorrhea - esp. with Risperidone)
- **Metabolic Syndrome** (Weight gain, Diabetes - esp. with Olanzapine, Clozapine)
- **Neuroleptic Malignant Syndrome (NMS)** – Life-threatening fever, rigidity → Treat with **Dantrolene**

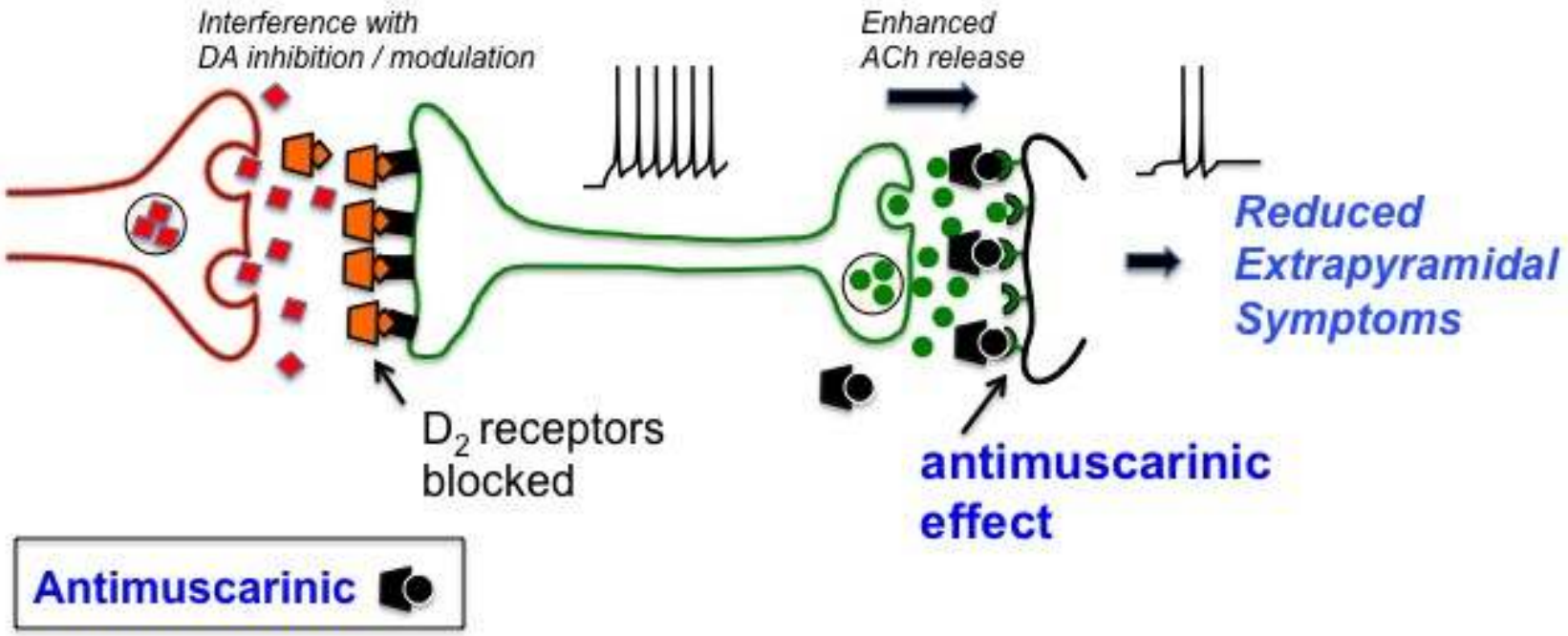
Adverse Effects of Antipsychotics



Cause of EPS and Treatment

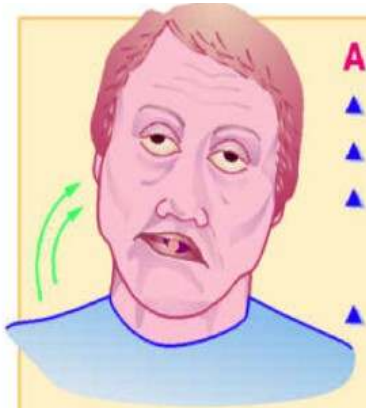
Extrapyramidal Syndrome (EPS) – due to dopamine receptor blockade of the nigro-striatal pathway

Treatment : antimuscarinics (benztropine, biperiden, trihexphenadyl, diphenhydramine) and switching to an antipsychotic having lower liability to cause EPS (aripiprazole)



Adverse effects: MOTOR DEFICITS increase with increase in duration of treatment – Mostly Due to classic or conventional Antipsychotics

Duration of Antipsychotic used	Adverse effects	Reversal of adverse effects
Hours to Days	Dystonias (neurological movement disorder: Spasms of muscles: tongue, face neck)	a. Diphenhydramine (Histamine and cholinergic blocker) or **b. Muscarinic blockers (Benztropine) IM or IV**
Days to Weeks	**Akathisias** (motor restlessness: Inability to remain seated)	a. Diphenhydramine or **b. Muscarinic blockers** **c. Benzodiazepines** **d. Beta blockers (Propranolol)**
Weeks to months	**Parkinson- Like symptoms** Bradykinesia, Tremors, Rigidity	**a. Decrease dose of antipsychotic** or b. Diphenhydramine or c. Muscarinic blocking agents
Months to years	***Tardive dyskinesia** facial-oral dyskinesia, dystonia, irregular spasmodic movements, athetosis	a. No effective drug treatment due to up regulation of dopaminergic system. b. Temporary effect



Acute dystonia

- Facial grimacing
- Involuntary upward eye movement
- Muscle spasms of the tongue, face, neck and back (back muscle spasms cause trunk to arch forward)
- Laryngeal spasms



Akathisia

- Restless
- Trouble standing still
- Paces the floor
- Feet in constant motion, rocking back and forth



Pseudoparkinsonism

- Stooped posture
- Shuffling gait
- Rigidity
- Bradykinesia
- Tremors at rest
- Pill-rolling motion of the hand



Tardive dyskinesia

- Protrusion and rolling of the tongue
- Sucking and smacking movements of the lips
- Chewing motion
- Facial dyskinesia
- Involuntary movements of the body and extremities

Other Adverse Effects

- **Neuroleptic malignant syndrome**
  - **Muscular rigidity, fever,** altered mental status and stupor, impairment of sweating, unstable blood pressure, and myoglobinemia
  - Rare condition
  - Treatment: withdraw drug immediately
  - Administration of **Dantrolene or Bromocriptine or diazepam or dopamine agonists** may be helpful

Indications of Antipsychotics

Psychiatric indications

1. Schizophrenia

- primary use
- used long term
- used with antidepressants, lithium, or valproic acid

2. Bipolar affective disorder

- For acute phase of mania only
- Olanzapine or Quetiapine

3. Gilles de la Tourette's **s**yndrome (TS)

4. Psychosis induced by Alzheimer's and Parkinson's disease

5. Toxic psychosis induced by CNS stimulants

Non-psychiatric indications

Antiemetic (typical antipsychotics)

- D_2 block: CTZ (medulla)
- chlorpromazine, haloperidol used at low doses

A 25-year-male was diagnosed as a case of schizophrenia on the basis of disturbed thinking process, inappropriate talking and behaviour, restlessness, bursts of temper, anxiety, poor selfcare, disturbed sleep, delusional beliefs and occasional auditory hallucinations. He was treated with:

Tab. Haloperidol 5 mg once daily at bed time.

The dose was increased to 7.5 mg daily in the 2nd week and to 10 mg daily in the 3rd week. His symptoms gradually subsided and he appeared more calm and organized. However, in the 5th week his family members reported that his restlessness has reappeared, he keeps pacing around in the room, but is not aggressive or combative. On questioning, the patient admitted an uncontrollable urge to move around and that he feels uncomfortable in remaining still. He is not worried or anxious, but has difficulty in falling asleep.

(a) What could be the reason for the motor restlessness? Should the dose of haloperidol be increased or decreased, or should it be changed to another antipsychotic drug?

(b) Should any other drug be given to relieve the condition?

THANK YOU